

Solution for Homework 1: Set Theory

1.26: To determine which sets are equal, let's first find the solutions for the given equations:

For set A: $x^2 - 4x + 3 = (x - 3)(x - 1) = 0$. Therefore, set A = {1, 3}.

For set B: $x^2 - 3x + 2 = (x - 2)(x - 1) = 0$. Therefore, set B = {1, 2}.

Now, let's compare these sets to the other sets:

$$C = \{x \mid x \in \mathbb{N}, x < 3\} = \{1, 2\}$$

$$E = \{1, 2\}$$

$$G = \{3, 1\}$$

$$D = \{x \mid x \in \mathbb{N}, x \text{ is odd}, x < 5\} = \{1, 3\}$$

$$F = \{1, 2, 1\} = \{1, 2\}$$

$$H = \{1, 1, 3\} = \{1, 3\}$$

Now, let's compare the sets:

$$A = B = \{1, 2\}$$

$$C = E = F = \{1, 2\}$$

$$G = D = H = \{1, 3\}$$

So, the sets that are equal are:

A and B

C, E, and F

G, D, and H

1.27 Sure, here are the elements of each set, and I will also identify which sets, if any, are equal:

Let $U = \{a, b, c, \dots, y, z\}$.

Set $A = \{x \mid x \text{ is a vowel}\}$: $A = \{a, e, i, o, u\}$.

Set $C = \{x \mid x \text{ precedes } f \text{ in the alphabet}\}$: $C = \{a, b, c, d, e\}$.

Set $B = \{x \mid x \text{ is a letter in the word "little"}\}$: $B = \{l, i, t, e\}$.

Set $D = \{x \mid x \text{ is a letter in the word "title"}\}$: $D = \{t, i, l, e\}$.

Thus $B = D$

1.29: (a) $A \cap B = \{2, 5\}$, $A \cap C = \{1, 5\}$

(b) $A \cup B = \{1, 2, 5, 6, 7\}$, $B \cup C = \{1, 2, 3, 5, 7, 9\}$

(c) $A^C = \{3, 4, 7, 8, 9\}$, $C^C = \{2, 4, 6, 8\}$

(d) $A \setminus B = \{1, 6\}$, $A \setminus C = \{2, 5, 6\}$

(e) $A \oplus B = \{1, 7\}$, $A \oplus C = \{2, 3, 7, 9\}$

(f) $(A \cup C) \setminus B = \{1, 3, 9\}$, $(B \oplus C) \setminus A = \{3, 4, 6, 8\}$

1.30: (a) To show that A is the disjoint union of $A \setminus B$ and $A \cap B$, we need to prove two things:

1. $A = (A \setminus B) \cup (A \cap B)$ and 2. $(A \setminus B) \cap (A \cap B) = \emptyset$ (i.e., the intersection is empty)

Proof of (1):

$$\begin{aligned}(A \setminus B) \cup (A \cap B) &= (A \cap B') \cup (A \cap B) \quad [\text{By set difference definition}] \\ &= A \cap (B' \cup B) \quad [\text{Distributive property}] \\ &= A \cap U \quad [\text{Complement law}] \\ &= A \quad [\text{Identity law}]\end{aligned}$$

Proof of (2):

$$\begin{aligned}
(A \setminus B) \cap (A \cap B) &= (A \cap B') \cap (A \cap B) \quad [\text{By set difference definition}] \\
&= A \cap (B' \cap B) \quad [\text{Distributive property}] \\
&= A \cap \emptyset \quad [\text{Complement law}] \\
&= \emptyset \quad [\text{Identity law}]
\end{aligned}$$

(b) To show that $A \cup B$ is the disjoint union of $A \setminus B$, $A \cap B$, and $B \setminus A$, we need to prove three things:

1. $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$ 2. $(A \setminus B) \cap (A \cap B) = \emptyset$ 3. $(A \setminus B) \cap (B \setminus A) = \emptyset$
Proof of (1):

$$\begin{aligned}
(A \setminus B) \cup (A \cap B) \cup (B \setminus A) \\
&= (A \cap B') \cup (A \cap B) \cup (B \cap A') \quad [\text{By set difference definition}] \\
&= A \cap (B' \cup B) \cup (B \cap A') \quad [\text{Distributive property}] \\
&= (A \cap U) \cup (B \cap A') \quad [\text{Complement law}] \\
&= A \cup (B \cap A') \quad [\text{Identity law}] \\
&= (A \cup B) \cap (A \cup A') \quad [\text{Distributive property}] \\
&= (A \cup B) \cap U \quad [\text{Complement law}] \\
&= (A \cup B) \quad [\text{Identity law}]
\end{aligned}$$

Proof of (2) and (3) are similar to part (a), just with different sets.

- 1.31: (a) $A \subseteq B$ if and only if $A \cap B' = \emptyset$.

Proof:

1. $A \subseteq B \Rightarrow A \cap B' = \emptyset$: $A \subseteq B \Rightarrow \forall x \in A, x \in B \Rightarrow x \notin B' \Rightarrow A \cap B' = \emptyset$
2. $A \cap B' = \emptyset \Rightarrow A \subseteq B$: $A \cap B' = \emptyset \Rightarrow \forall x \in A, x \notin B' \Rightarrow x \in B \Rightarrow A \subseteq B$

- (b) $A \subseteq B$ if and only if $A' \cup B = U$.

Proof:

1. $A \subseteq B \Rightarrow A' \cup B = U$: $A \subseteq B \Rightarrow \forall x \in A, x \in B \Rightarrow x \notin B' \Rightarrow A \cap B' = \emptyset \Rightarrow \underbrace{(A \cap B')' = U}_{\text{De Morgan}} \Rightarrow A' \cup B = U$
2. $A' \cup B = U \Rightarrow A \subseteq B$: $(A' \cup B) = U \Rightarrow \underbrace{(A' \cup B)' = \emptyset}_{\text{De Morgan's \& from (a) above}} \Rightarrow (A \cap B') = \emptyset \Rightarrow A \subseteq B$

- (c) $A \subseteq B$ if and only if $B' \subseteq A'$

Proof:

1. $A \subseteq B \Rightarrow B' \subseteq A'$: Let $x \in B' \Rightarrow \underbrace{x \notin B \Rightarrow x \notin A}_{A \subseteq B} \Rightarrow x \in A' \Rightarrow B' \subseteq A'$
2. $B' \subseteq A' \Rightarrow A \subseteq B$: Let $y \in A \Rightarrow y \notin A' \Rightarrow y \notin B' \Rightarrow y \in B \Rightarrow A \subseteq B$

- (d) $A \subseteq B$ if and only if $A \setminus B = \emptyset$. Proof: Like in (a) above:

1.34:

1.37: Dual Equation (a):

$$A' = (B \cup A') \cap (A' \cup B')$$

Dual Equation (b):

$$(A' \cup B') \cap (A \cup B') \cap (A' \cup B) \cap (A \cup B) = \emptyset$$

- 1.39: (a) Lines parallel to the x-axis: This set is not finite. There are infinitely many lines parallel to the x-axis, as they can have any real number as their y-coordinate.
- (b) Letters in the English alphabet: This set is finite. There are 26 letters in the English alphabet: $\{A, B, C, \dots, Z\}$.
- (c) Integers which are multiples of 5: This set is infinite. There are infinitely many integers that are multiples of 5 (e.g., -10, -5, 0, 5, 10, ...).
- (d) Animals living on the earth: This set is not finite. There are countless species of animals on Earth, and new species continue to be discovered. The set of all animals living on Earth is vast and not finite.

1.40:

$$\begin{aligned}
 n(A \cup B \cup C) &= n((A \cup B) \cup C) \\
 &= n(A \cup B) + n(C) - n((A \cup B) \cap C) \\
 &= n(A) + n(B) + n(C) - n(A \cap B) - n((A \cap C) \cup (B \cap C)) \\
 &= n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)
 \end{aligned}$$

- 1.41: To find the number of cars that had specific combinations of options (a) through (h), we can use the principle of inclusion-exclusion and the given information.

Let's denote: - $n(A)$ as the number of cars with air conditioning.

- $n(R)$ as the number of cars with a radio.

- $n(W)$ as the number of cars with power windows.

- $n(A \cap R)$ as the number of cars with both air conditioning and a radio.

- $n(A \cap W)$ as the number of cars with both air conditioning and power windows.

- $n(R \cap W)$ as the number of cars with both a radio and power windows.

- $n(A \cap R \cap W)$ as the number of cars with all three options.

Given information:

- $n(A) = 15$

- $n(A \cap W) = 5$

- $n(R) = 12$

- $n(A \cap R) = 9$

- $n(A \cap R \cap W) = 3$

- $n(W) = 11$

- $n(R \cap W) = 4$

Now, let's find the number of cars with each of the specified combinations:

(a) Only power windows (W): $n(W) - n(A \cap W) - n(R \cap W) + n(A \cap R \cap W) = 11 - 5 - 4 + 3 = 5$

(b) Only air conditioning (A): $n(A) - n(A \cap R) - n(A \cap W) + n(A \cap R \cap W) = 15 - 9 - 5 + 3 = 4$

(c) Only radio (R): $n(R) - n(A \cap R) - n(R \cap W) + n(A \cap R \cap W) = 12 - 9 - 4 + 3 = 2$

(d) Radio and power windows but not air conditioning (R and W but not A): $n(R \cap W) - n(A \cap R \cap W) = 4 - 3 = 1$ car.

(e) Air conditioning and radio but not power windows (A and R but not W): $n(A \cap R) - n(A \cap R \cap W) = 9 - 3 = 6$ cars.

(f) Only one of the options (exclusive OR): This is the sum of cars with only A, only R, and only W: $4 + 2 + 5 = 11$ cars.

(g) At least one option: This is the complement of cars with none of the options, which we will calculate in (h).

(h) None of the options:

$$n(\text{None}) = \text{Total Cars} - (n(A) + n(R) + n(W) - n(A \cap R) - n(A \cap W) - n(R \cap W) + n(A \cap R \cap W))$$

$$= 25 - (15 + 12 + 11 - 9 - 5 - 4 + 3) = 7$$

cars.

$$1.42: P(A) = \left\{ \begin{array}{l} \emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \\ \{3, 4\}, \{3, 5\}, \{4, 5\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{1, 3, 4\}, \{1, 3, 5\}, \{1, 4, 5\}, \\ \{2, 3, 4\}, \{2, 3, 5\}, \{2, 4, 5\}, \{3, 4, 5\}, \{1, 2, 3, 4\}, \{1, 2, 3, 5\}, \{1, 2, 4, 5\}, \\ \{1, 3, 4, 5\}, \{2, 3, 4, 5\}, \{1, 2, 3, 4, 5\} \end{array} \right\}$$

1.44: To prove that the power set $P(A)$ of a finite set A with $n(A) = m$ has 2^m elements, you can use the concept of counting subsets.

1. Each element in the set A has two choices: either it is included in a subset or it is not. Therefore, for each element of A , there are 2 possibilities.

2. Since there are m elements in set A , the total number of possible subsets is the product of the number of choices for each element, which is 2^m .

3. Therefore, $P(A)$ has 2^m elements.

This proves that the power set $P(A)$ of a finite set A with $n(A) = m$ contains 2^m elements.

1.45: To be considered a partition of $S = \{1, 2, \dots, 8, 9\}$, the sets within the partition must satisfy two conditions:

1. Non-emptiness: Each set in the partition must be non-empty.

2. Mutual Exclusivity: The sets in the partition must be mutually exclusive, meaning no element appears in more than one set.

Let's evaluate each of the provided partitions:

(a) $\{\{1, 3, 6\}, \{2, 8\}, \{5, 7, 9\}\}$ - Non-emptiness: Each set is non-empty. - Mutual Exclusivity: The sets are mutually exclusive, no element appears in more than one set.

This partition satisfies both conditions, so it is a valid partition of S .

(b) $\{\{1, 5, 7\}, \{2, 4, 8, 9\}, \{3, 5, 6\}\}$ - Non-emptiness: Each set is non-empty. - Mutual Exclusivity: The sets are mutually exclusive, no element appears in more than one set.

This partition also satisfies both conditions, so it is a valid partition of S .

(c) $\{\{2, 4, 5, 8\}, \{1, 9\}, \{3, 6, 7\}\}$ - Non-emptiness: Each set is non-empty. - Mutual Exclusivity: The sets are mutually exclusive, no element appears in more than one set.

This partition satisfies both conditions, so it is a valid partition of S .

(d) $\{\{1, 2, 7\}, \{3, 5\}, \{4, 6, 8, 9\}, \{3, 5\}\}$ - Non-emptiness: Each set is non-empty. - Mutual Exclusivity: The sets are not mutually exclusive. Set $\{3, 5\}$ appears twice in the partition, violating the condition.

This partition does not satisfy the mutual exclusivity condition, so it is not a valid partition of S .

1.47: Let's evaluate each of the provided partitions:

(a) $5 \notin [\{n \mid n > 5\}, \{n \mid n < 5\}]$ Not

(b) $6 \notin [\{n \mid n > 6\}, \{1, 3, 5\}, \{2, 4\}]$ Not

(c) $[\{n \mid n^2 > 11\}, \{n \mid n^2 < 11\}] = [\{1, 2, 3\}, \{4, 5, \dots\}]$ Yes

1.48: To show that the collection of sets $P = \{A_i \cap B_j | i = 1, \dots, m, j = 1, \dots, n\} \setminus \{\emptyset\}$ is a partition of the set S , we need to prove that

- (a) Every element of S belongs to at least one set in P .
- (b) The sets in P are pairwise disjoint.

Let's prove both of these properties:

Every element of S belongs to at least one set in P :

(a) For any $x \in S$, $x \in A_k; 1 \leq k \leq m$ and $x \in B_l; 1 \leq l \leq n$. Therefore, $x \in A_k \cap B_l$, and by the definition of P , x also belongs to $A_i \cap B_j$. So, x belongs to at least one set in P .

(b) The sets in P are pairwise disjoint:

Let $(A_r \cap B_s), (A_k \cap B_l) \in P$ with $A_r \cap B_s \neq \emptyset, A_k \cap B_l \neq \emptyset$ assuming that $(A_r \cap B_s) \cap (A_k \cap B_l) \neq \emptyset$, then there is some $y \in S$ such that

$$y \in (A_r \cap B_s) \cap (A_k \cap B_l) = (A_r \cap A_k) \cap (B_s \cap B_l)$$

This implies that $A_r \cap A_k = \emptyset$ and $B_s \cap B_l = \emptyset$ which is a contradiction. Thus $(A_r \cap B_s) \cap (A_k \cap B_l) = \emptyset$. Hence P is a partition of S

1.49: To find the cross partition P of the given partitions P_1 and P_2 of the set $S = \{1, 2, 3, \dots, 8, 9\}$, we'll compute the intersections of the sets in P_1 and P_2 following the formula for $P = \{A_i \cap B_j | i = 1, \dots, m, j = 1, \dots, n\} \setminus \{\emptyset\}$.

Given:

$$P_1 = [\{1, 3, 5, 7, 9\}, \{2, 4, 6, 8\}]$$

$$P_2 = [\{1, 2, 3, 4\}, \{5, 7\}, \{6, 8, 9\}]$$

Let's find the intersections of sets in P_1 and P_2 and then eliminate the empty set ($\emptyset$):

$$P_1 \cap P_2:$$

$$- P_1 \cap P_2 = \{\{1, 3, 5, 7, 9\} \cap \{1, 2, 3, 4\}, \{2, 4, 6, 8\} \cap \{1, 2, 3, 4\}, \{1, 3, 5, 7, 9\} \cap \{5, 7\}, \{2, 4, 6, 8\} \cap \{5, 7\}, \{1, 3, 5, 7, 9\} \cap \{6, 8, 9\}, \{2, 4, 6, 8\} \cap \{6, 8, 9\}\}$$

Compute the intersections:

- $\{1, 3, 5, 7, 9\} \cap \{1, 2, 3, 4\} = \{1, 3\}$
- $\{2, 4, 6, 8\} \cap \{1, 2, 3, 4\} = \{2, 4\}$
- $\{1, 3, 5, 7, 9\} \cap \{5, 7\} = \{5, 7\}$
- $\{2, 4, 6, 8\} \cap \{5, 7\} = \{\}$ (empty set)
- $\{1, 3, 5, 7, 9\} \cap \{6, 8, 9\} = \{9\}$
- $\{2, 4, 6, 8\} \cap \{6, 8, 9\} = \{6, 8\}$

Eliminate the empty set:

$$- P_1 \cap P_2 = \{\{1, 3\}, \{2, 4\}, \{5, 7\}, \{9\}, \{6, 8\}\}$$

So, the cross partition P of the given partitions P_1 and P_2 is:

$$P = \{\{1, 3\}, \{2, 4\}, \{5, 7\}, \{9\}, \{6, 8\}\}$$